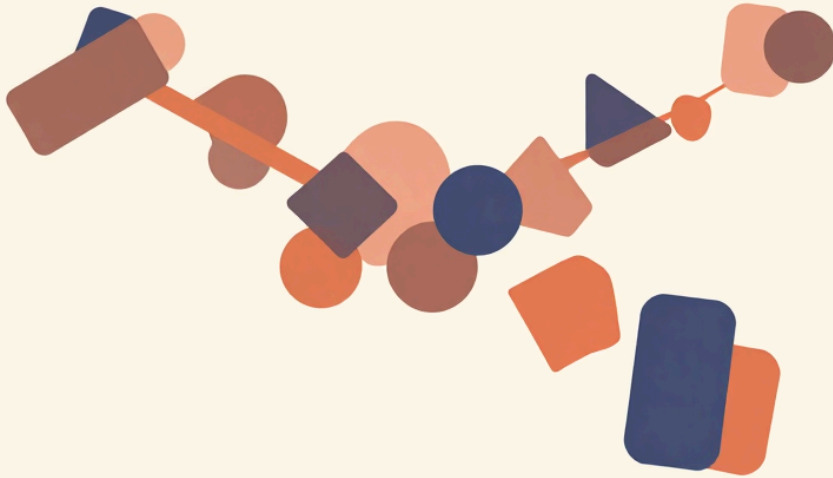


GameHorizon Suite: Multi-Horizon Data and Evaluation in Gameplay



PAPER EXPLAINED

Why AI agents fail at AAA games

GameHorizon Suite: Multi-Horizon Data and Evaluation in Gameplay

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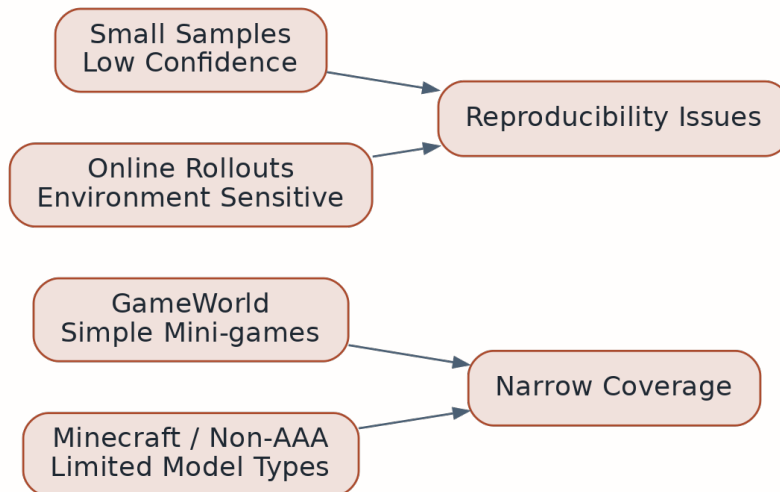
<https://arxiv.org/abs/2609.25001>

Measuring Multi-Horizon Gameplay

- The paper introduces GameHorizon, a unified data and evaluation framework.
- The corpus comprises 5,000 hours of human gameplay across 21 game titles.
- The suite evaluates long-horizon gameplay through order-dependent causal tasks and order-flexible thematic tasks.
- Each task contains 2-6 verifiable subtasks to assess short-horizon capabilities.

Narrow Coverage and Reproducibility Issues

- Existing datasets like GameWorld target only simple mini-games, resulting in narrow game coverage.
- Prior methods use small samples for low-confidence comparisons and online rollouts that are hard to reproduce.
- Earlier work focuses on Minecraft or non-AAA games.



Limitations of prior game benchmarks: narrow coverage and reproducibility issues.

Annotator Pipeline and Benchmark Design

- Action-aware segmentation uses keyboard-mouse traces to identify key action transitions and determine clip boundaries.
- Dynamic programming enforces duration ranges: 1–5 seconds for L1, 1–2 minutes for L2, and 5–8 minutes for L3.
- Unified multiple-choice questions ensure standardized evaluation where models directly output selections from four options.



Annotator pipeline: from raw traces to standardized benchmark clips and questions.

Proprietary Models Lead, UMMs Lag

- Mean accuracy is 64.7%, substantially above the 25% random baseline.
- Proprietary model GPT-6-Astra ranks first with 80.2% overall accuracy.
- Best UMM BAGEL-7B-MoT achieves 61.4%, falling into Tier 3 or Tier 4.



Proprietary model GPT-6-Astra leads with 80.2% accuracy, while UMMs lag behind the mean of 64.7%.